

Cold Storage Commodity and Chamber Management System

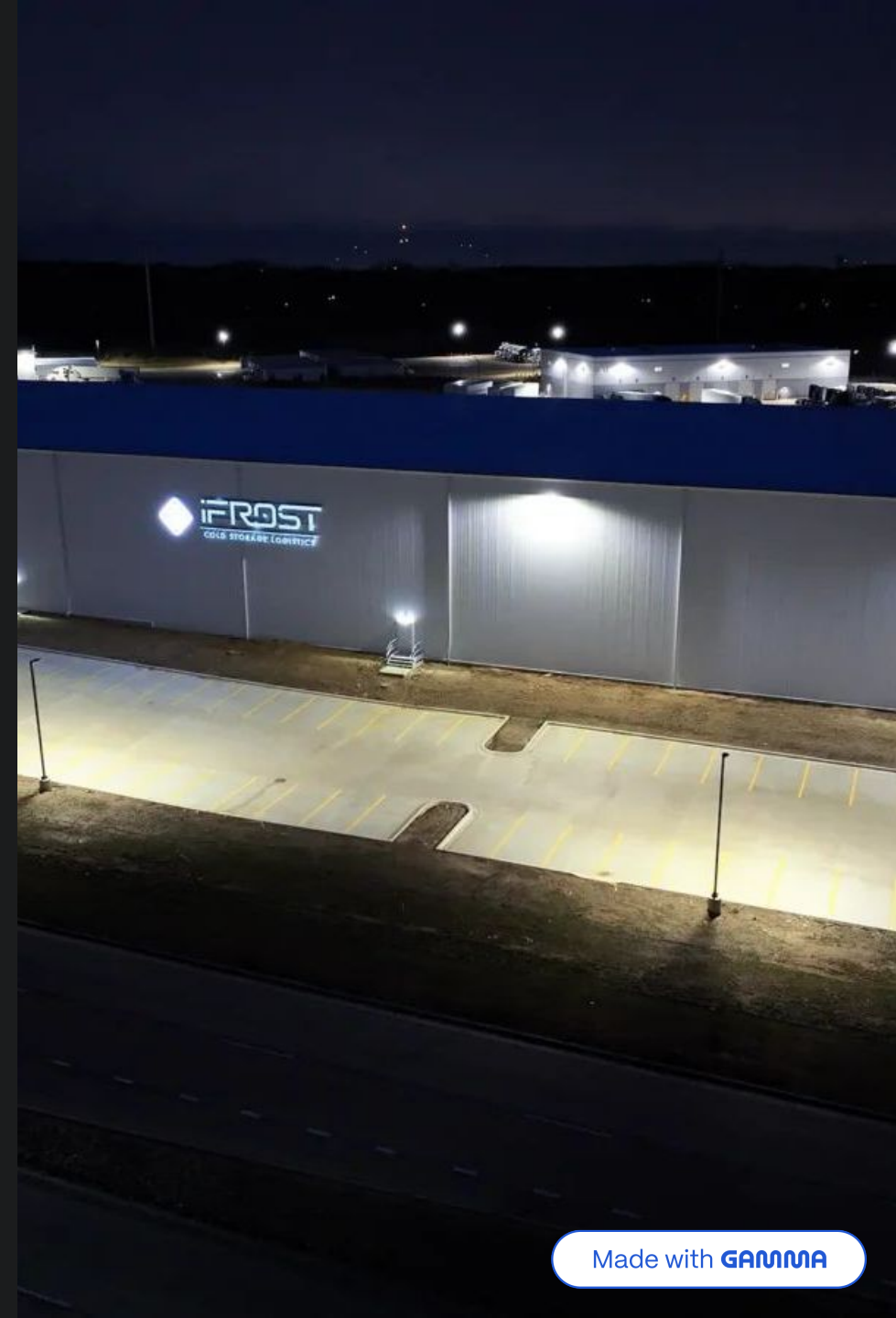
DBMS Course Project — Review 2

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Project Overview

Purpose: Implement a relational database to manage cold-storage operations using the supplied SQL schema and sample data.



Customers

Customer master data and contact details.



Commodities & Lots

Commodity definitions, lots with arrival, quantity and status.



Chambers, Locations, Allocations

Physical storage (chambers → locations) and lot allocations.



Monitoring

Temperature logs and inspections tied to allocations.



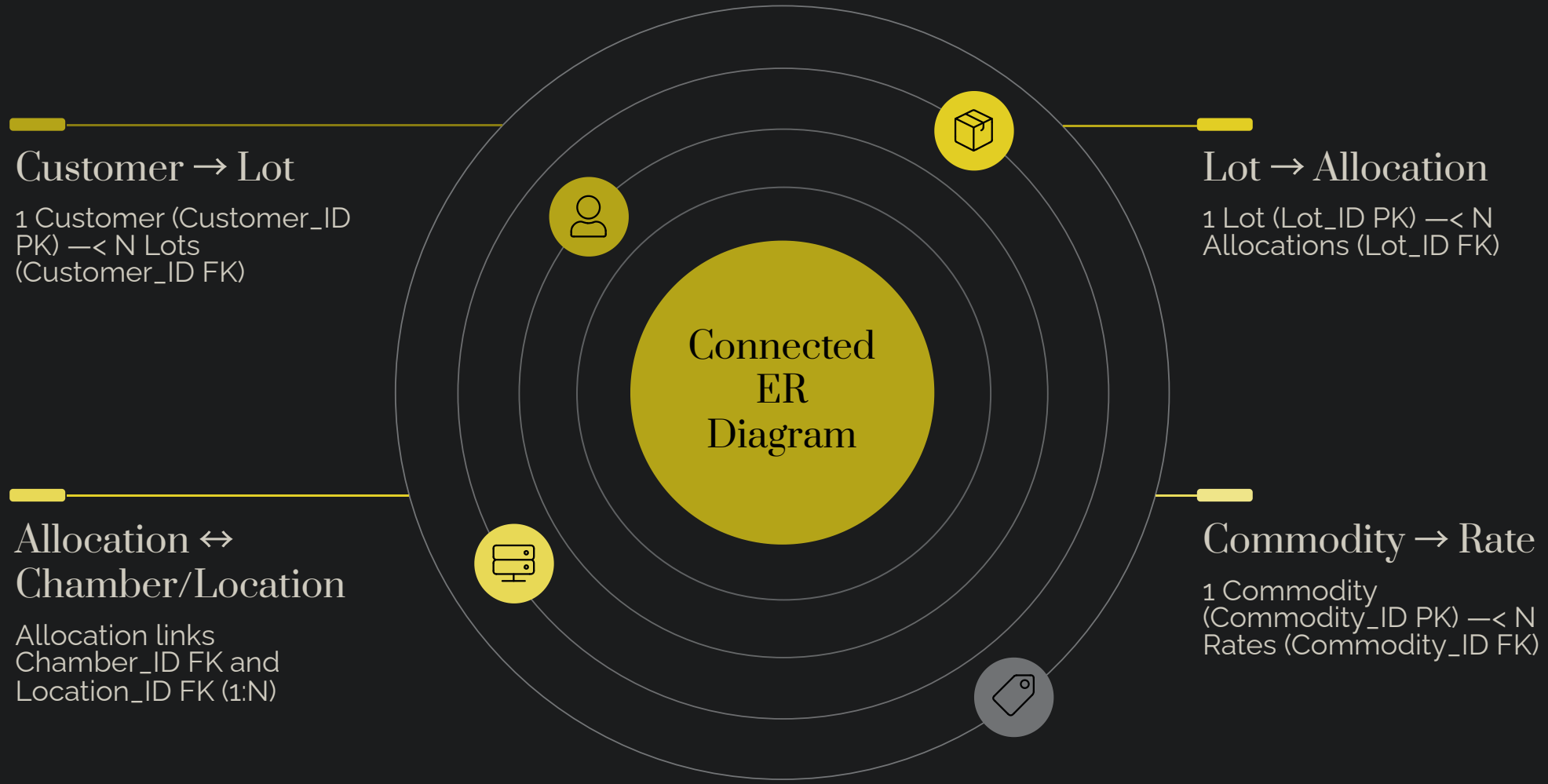
Billing

Rates, releases, invoices and payments to settle storage charges.

Flow (FK-driven): Customer → Lot → Allocation → (Chamber, Location) → Monitoring/Inspection → Release → Invoice → Payment. Rates link Commodity → Rate.

ER Diagram (derived from SQL)

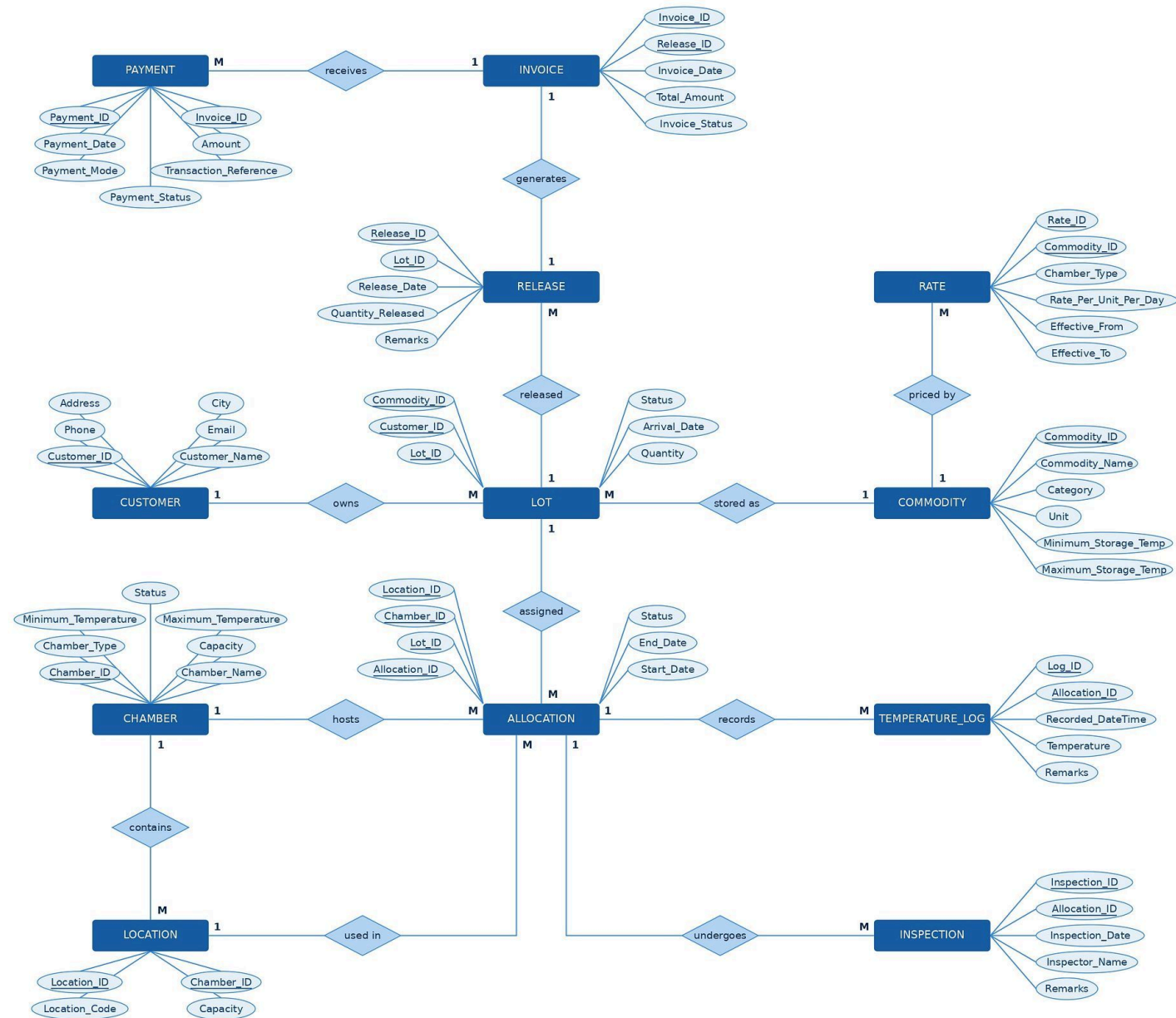
Entities, primary keys, foreign keys, key attributes and cardinalities are shown below. Relationships are derived strictly from implemented foreign keys.



Note: All relationships map directly to the SQL foreign keys; no extra relationships added.

Cold storage commodity and chamber management system

Underlined = primary key ~~Dashed~~ = foreign key 1 / M = cardinality



Database Schema — Core Tables (1/2)

Customer

Customer_ID — PK
Customer_Name, Phone, Email,
Address, City

Commodity

Commodity_ID — PK
Commodity_Name (UNIQUE),
Category, Unit,
Minimum_Storage_Temperature,
Maximum_Storage_Temperature

Chamber

Chamber_ID — PK
Chamber_Name (UNIQUE),
Chamber_Type, Capacity,
Minimum_Temperature,
Maximum_Temperature, Status

Rate

Rate_ID — PK
Commodity_ID — FK,
Chamber_Type,
Rate_Per_Unit_Per_Day,
Effective_From, Effective_To

Book

isbn	varchar(50)
title	varchar(100)
pages	integer
price	money
description	varchar(256)
publisher	varchar(100)

Book_details

id	integer
book_isbn	varchar(50)
rating	decimal
language	varchar(10)
keywords	text
date_publisher	date



Database Schema — Storage Tables (2/2)

Lot

Lot_ID — PK
Customer_ID — FK,
Commodity_ID — FK, Quantity,
Arrival_Date, Status

Location

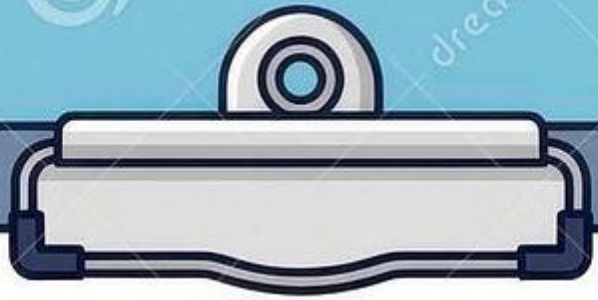
Location_ID — PK
Chamber_ID — FK,
Location_Code, Capacity

Allocation

Allocation_ID — PK
Lot_ID — FK, Chamber_ID —
FK, Location_ID — FK,
Start_Date, End_Date, Status

Temperature_Log

Log_ID — PK
Allocation_ID — FK,
Recorded_DateTime,
Temperature, Remarks



Cold Safety Checklist

- ☐ Dress in layers
- ☐ Stay hydrated
- ☐ Protect exposed skin
- ☒ Know frostbite signs
- ☐ Avoid alcohol
- ☒ Check vehicle fluids

Monitoring & Billing Tables

Monitoring

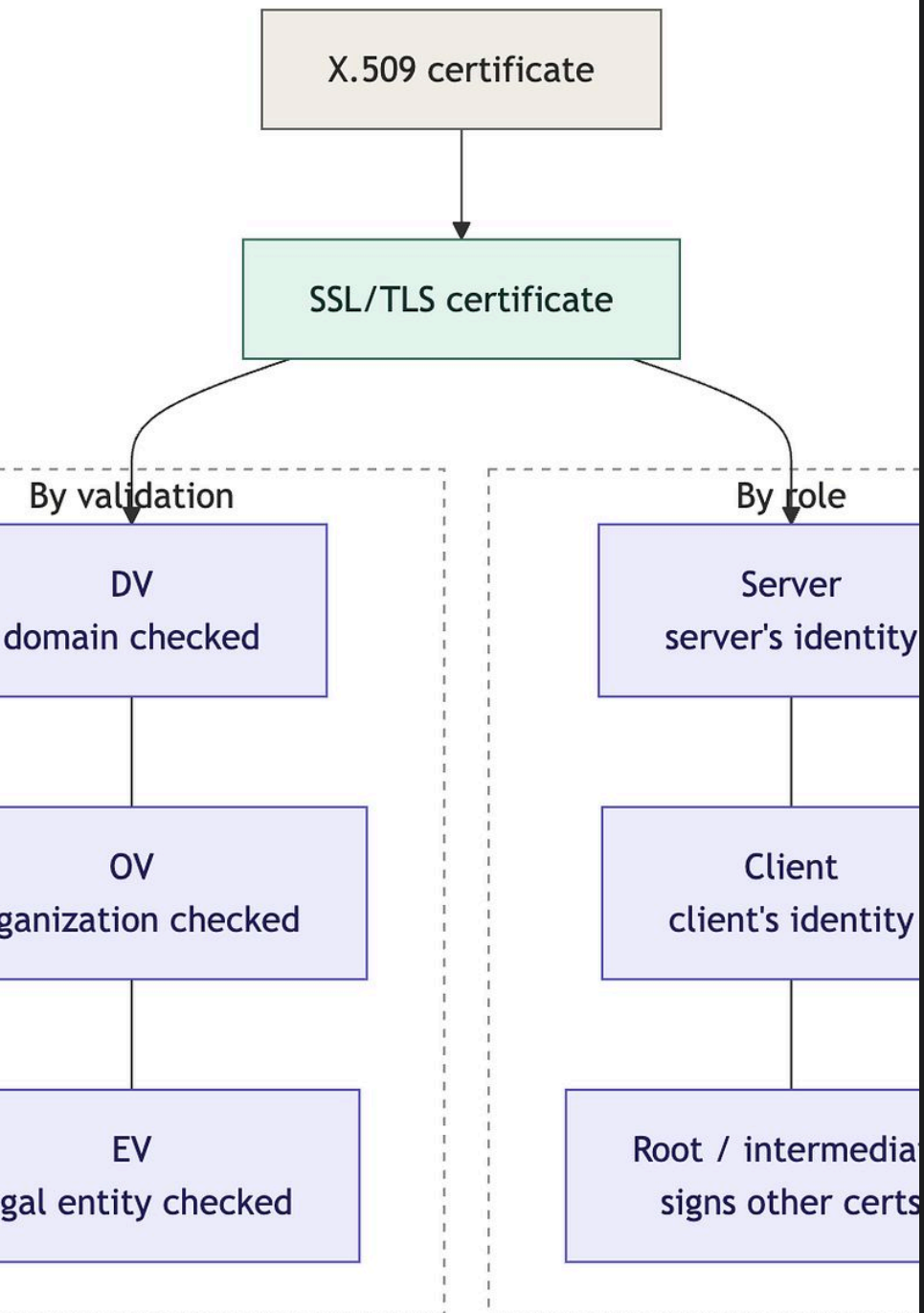
Inspection: Inspection_ID — PK, Allocation_ID — FK, Inspection_Date, Inspector_Name, Condition, Remarks

Release: Release_ID — PK, Lot_ID — FK, Release_Date, Quantity_Released, Remarks

Billing

Invoice: Invoice_ID — PK, Release_ID — FK, Invoice_Date, Total_Amount, Invoice_Status

Payment: Payment_ID — PK, Invoice_ID — FK, Payment_Date, Amount, Payment_Mode, Transaction_Reference, Payment_Status



Constraints & Data Integrity

Primary Keys

Every table implements a PK
(Customer_ID, Lot_ID, Allocation_ID,
etc.).

Foreign Keys

FKs enforce relationships:
Lot.Customer_ID → Customer,
Allocation.Lot_ID → Lot,
Invoice.Release_ID → Release,
Payment.Invoice_ID → Invoice, etc.

NOT NULL & UNIQUE

NOT NULL examples:
Customer_Name, Phone, City,
Commodity_Name, Quantity,
Arrival_Date. UNIQUE examples:
Commodity_Name, Chamber_Name,
(Chamber_ID, Location_Code),
Release_ID in Invoice.

CHECK Constraints

CHECK (Quantity > 0), CHECK
(Capacity > 0), CHECK
(Minimum_Storage_Temperature <
Maximum_Storage_Temperature),
CHECK (Rate_Per_Unit_Per_Day > 0),
CHECK (Amount > 0).

These constraints ensure referential integrity, valid numeric ranges, and uniqueness where required.

Filled-in Tables — Customer & Commodity

Customer (sample rows)

Customer_ID	Customer_Name	City
1	Ravi Traders	Nagpur
2	Kumar Foods	Pune
3	Sri Lakshmi Enterprises	Hyderabad
4	Green Valley Farms	Nashik
5	Fresh Harvest Co	Mumbai

Commodity (sample rows)

Commodity_ID	Commodity_Name	Category	Unit	Min Temp	Max Temp
1	Apples	Fruit	Kg	0	4
2	Mangoes	Fruit	Kg	10	13
3	Potatoes	Vegetable	Kg	4	10
4	Frozen Peas	Frozen	Kg	-18	-15
5	Grapes	Fruit	Kg	0	2

Filled-in Tables — Storage

Lot (selected)

Lot_ID	Customer_ID	Commodity_ID	Quantity	Arrival_Date / Status
L001	1	1	500	2024-01-10 / Stored
L002	2	3	800	2024-02-05 / Stored
L003	3	4	1200	2024-02-20 / Stored
L004	4	2	300	2024-03-01 / Stored

Chamber & Location (selected)

Chamber_ID	Chamber_Name	Location_Code	Capacity
CH01	CH01	RACK-A01	1000
CH02	CH02	RACK-A02	1000
CH03	CH03	RACK-B01	1500
CH04	CH04	RACK-F01	800

Allocation (selected)

Allocation_ID	Lot_ID	Chamber_ID	Location_ID	Status
A001	L001	CH01	RACK-A01	Active
A002	L002	CH02	RACK-A02	Active
A003	L003	CH03	RACK-B01	Active
A004	L004	CH04	RACK-F01	Active

Filled-in Tables — Monitoring & Billing

Temperature_Log (examples)

Log_ID	Allocation_ID	Recorded_DateTime	Temperature / Remarks
T001	A001	2024-01-11 06:00	2.5°C / Normal
T002	A001	2024-01-11 12:00	5.5°C / Above max — flagged
T003	A003	2024-02-21 09:00	-16.8°C / Normal

Inspection, Release, Invoice, Payment (selected)

Inspection_ID	Allocation_ID	Date	Inspector	Condition
INSP01	A001	2024-01-12	Mr. Sharma	Good

Release_ID	Lot_ID	Release_Date	Quantity_Released	Remarks
R001	L001	2024-02-01	200	Partial release

Invoice_ID	Release_ID	Invoice_Date	Total_Amount	Invoice_Status
INV001	R001	2024-02-02	₹12,000	Unpaid

Payment_ID	Invoice_ID	Payment_Date	Amount	Payment_Status
PAY001	INV001	2024-02-10	₹12,000	Completed